Assignment -02

Course Details

Course Title: Computer Networks

Couse Code: CSE 405

Section: 05

Submitted To

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Submitted By

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Introduction

This report describes the design and configuration of a four-router network created in Cisco Packet Tracer. The network consists of four routers (R0, R1, R2 and R3) connected in a serial DCE configuration, where each router is also connected to its own local network through copper straight-through cables. All IP addresses used in the design belong to Class C networks with a subnet mask of 255.255.255.0.

Static routes are manually configured on every router so that all devices within the topology can communicate seamlessly with one another. This setup demonstrates a clear understanding of IP addressing, inter-router connectivity and static route configuration all of which are fundamental concepts in computer networking. The figure is shown below:

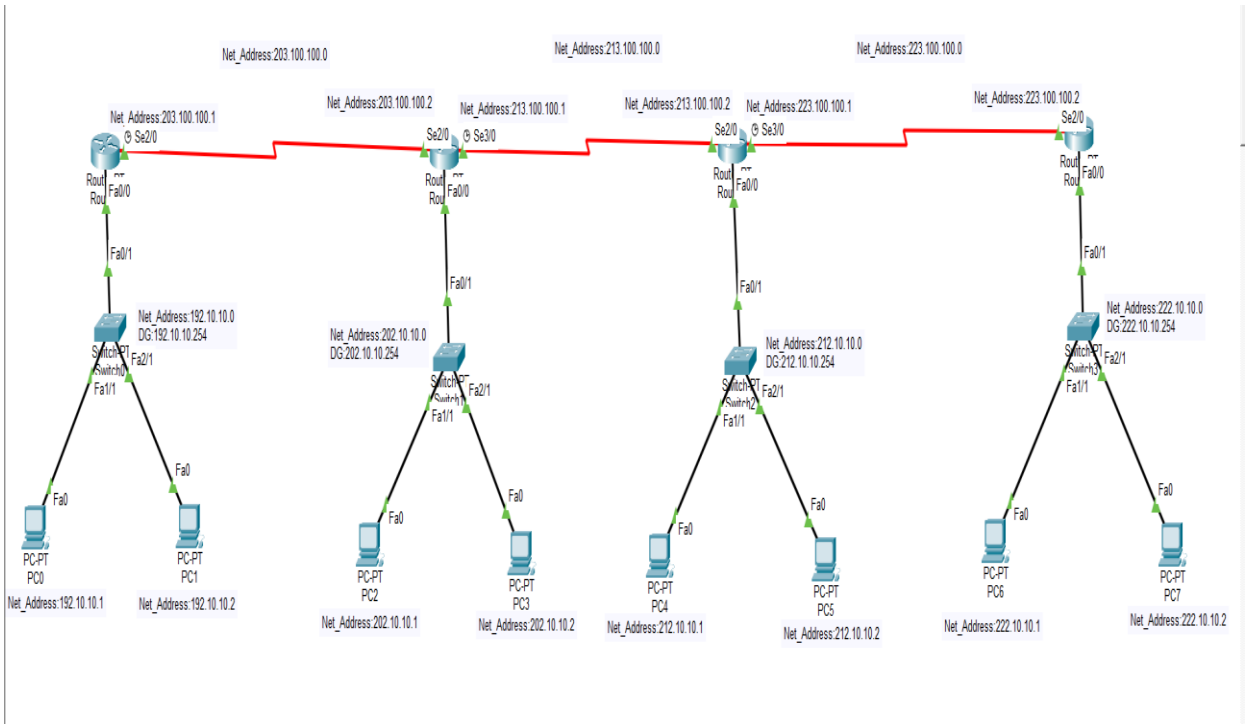


Figure: Router Configuration

Per device IP Address

- **R0**
  - Fa0/0: 192.10.10.254
  - Se2/0: 203.100.100.1
- **R1**
  - Fa0/0: 202.10.10.254
  - Se2/0: 203.100.100.2
  - Se3/0: 213.100.100.1
- **R2**

- Fa0/0: 212.10.10.254
- Se2/0: 213.100.100.2
- Se3/0: 223.100.100.1

- **R3**
  - Fa0/0: 222.10.10.254
  - Se2/0: 223.100.100.2

Host addresses

- PC0 = `192.10.10.1`, PC1 = `192.10.10.2`, Default gateway `192.10.10.254`
- PC2 = `202.10.10.1`, PC3 = `202.10.10.2`, `Default` gateway `202.10.10.254`
- PC4 = `212.10.10.1`, PC5 = `212.10.10.2`, Default gateway `212.10.10.254`
- PC6 = `222.10.10.1`, PC7= `222.10.10.2`, `Default` gateway `222.10.10.254`

Router CLI configurations (full commands)

For Router 0:

Fa0/0

```
Router>en
Router#config t
Router(config)#int fa0/0
Router(config-if)#ip address 192.10.10.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit
```

se2/0

```
Router(config)#int se2/0
Router(config-if)#ip address 203.100.100.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
Router(config)#exit
```

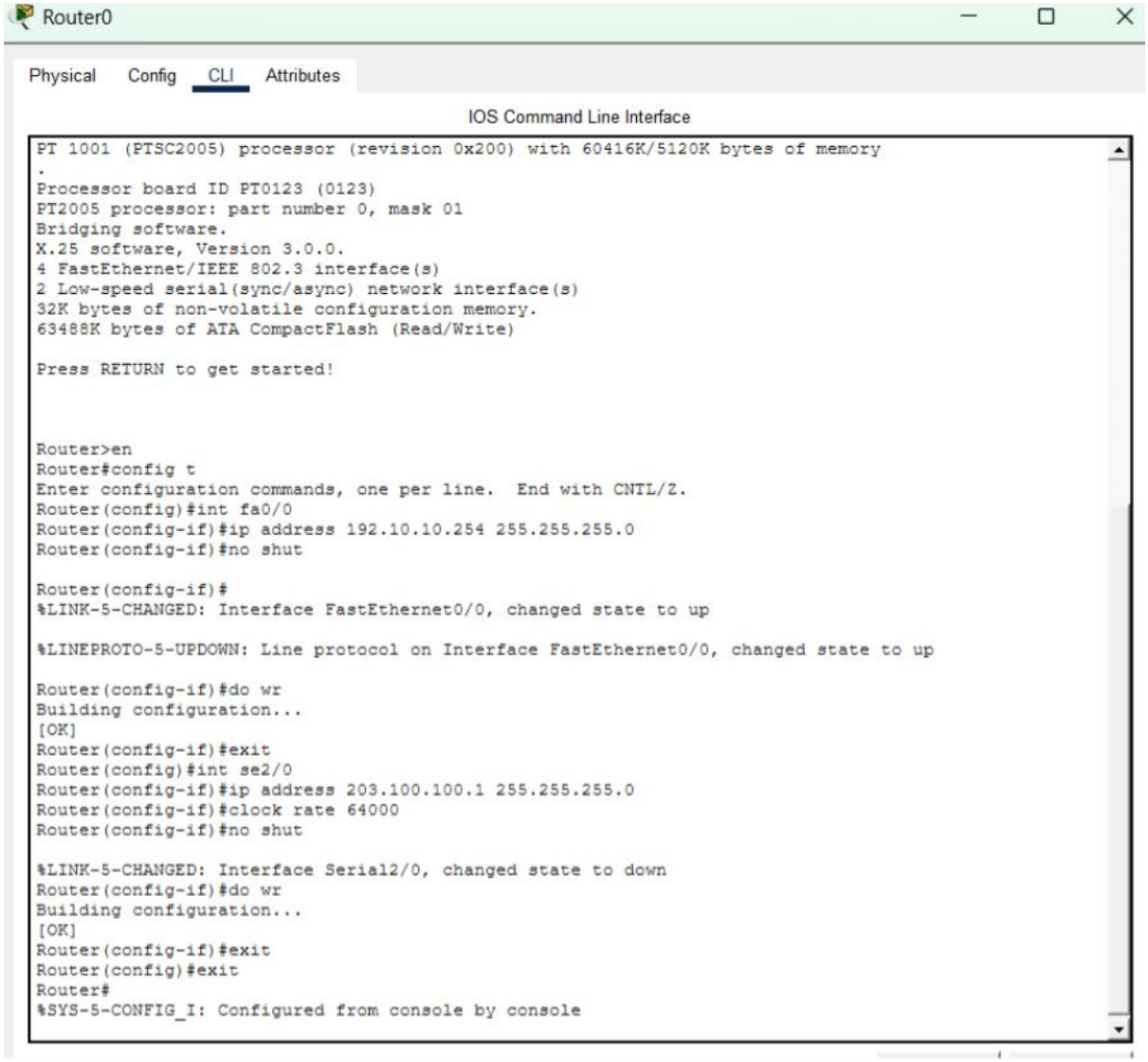


Figure : Router0 configuration

For Router 1:

se2/0

Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 203.100.100.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#
Router(config-if)#do wr
Router(config-if)#exit

fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 202.10.10.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

se3/0

Router(config)#int se3/0
Router(config-if)#ip address 213.100.100.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
Router(config)#exit

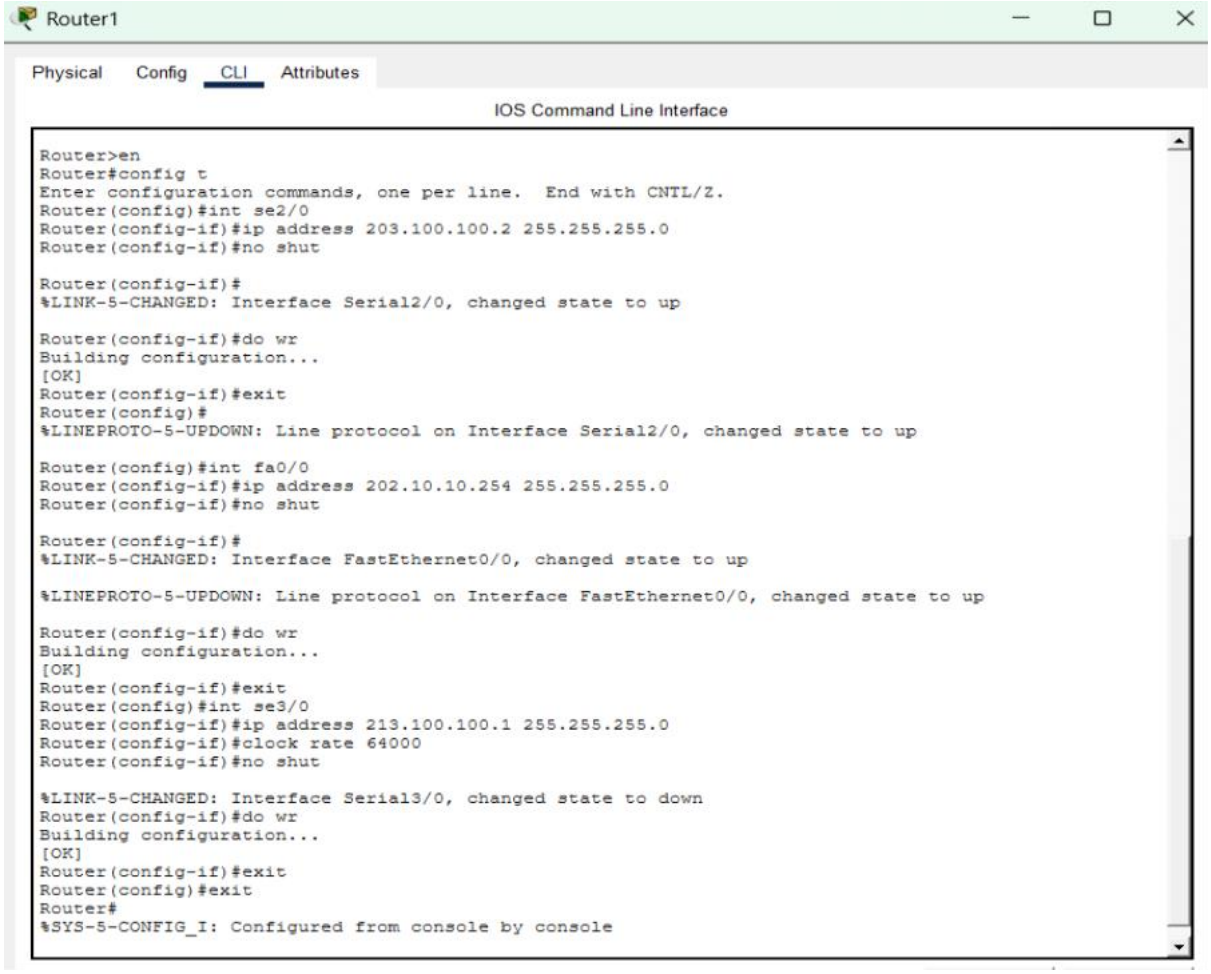


Figure : Router1 configuration

For Router 02:

se2/0

```
Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 213.100.100.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 212.10.10.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 223.100.100.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
Router(config)#exit
```

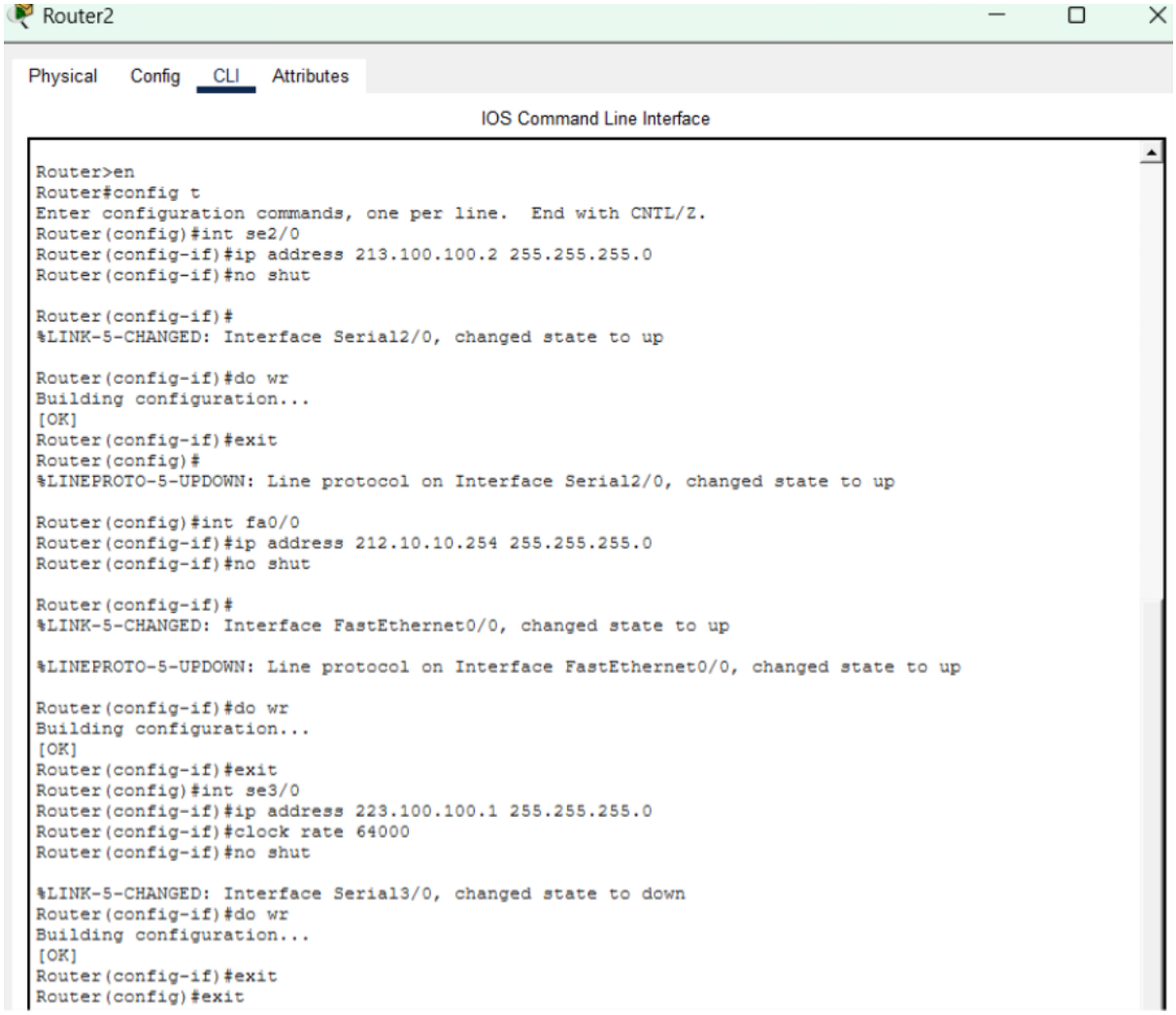


Figure: Router2 configuration

For Router 3:

se2/0

Router>en
Router#config t
Router(config)#int se2/0
Router(config-if)#ip address 223.100.100.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 222.10.10.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#do wr
Router(config-if)#exit
Router(config)#exit

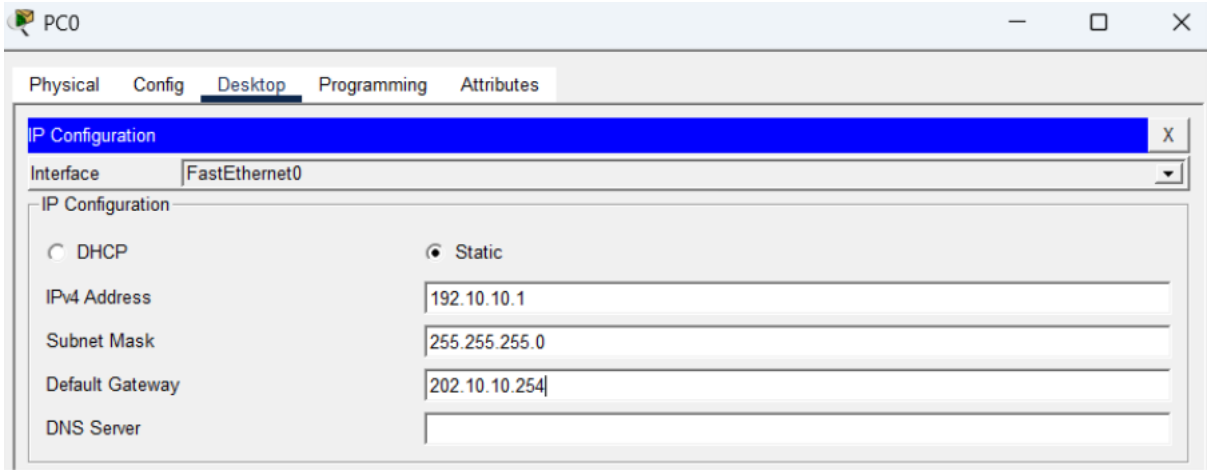
Physical Config CLI Attributes

IOS Command Line Interface

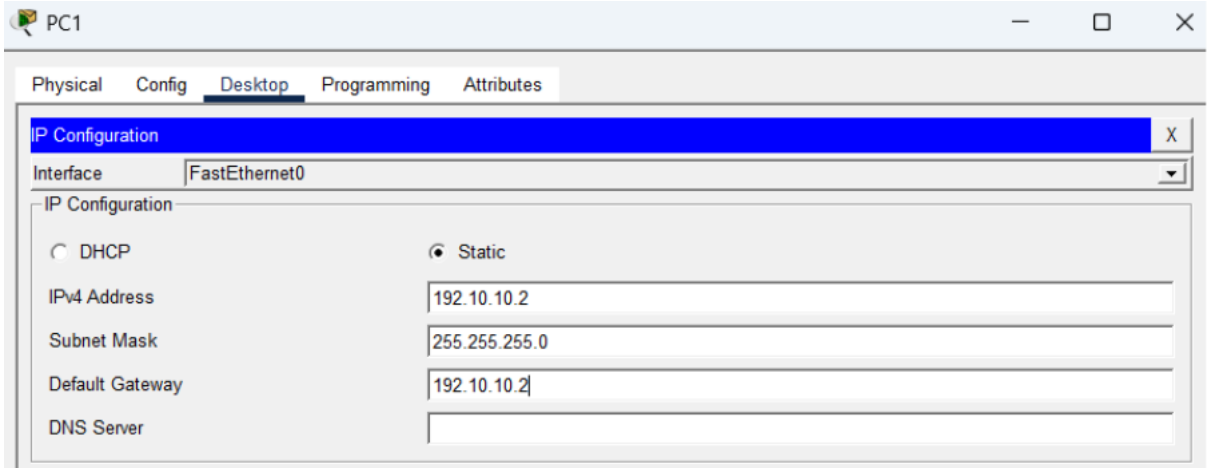
```
Bridging software.
X.25 software, Version 3.0.0.
4 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

Press RETURN to get started!



Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se2/0
Router(config-if)#ip address 223.100.100.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 222.10.10.254 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#exit
```

Figure : Router3 configuration

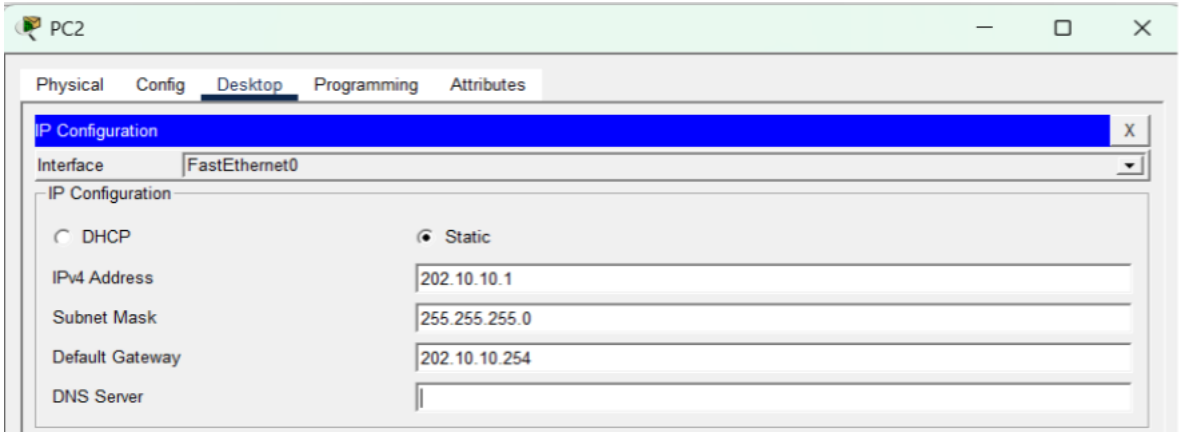
For PC 0



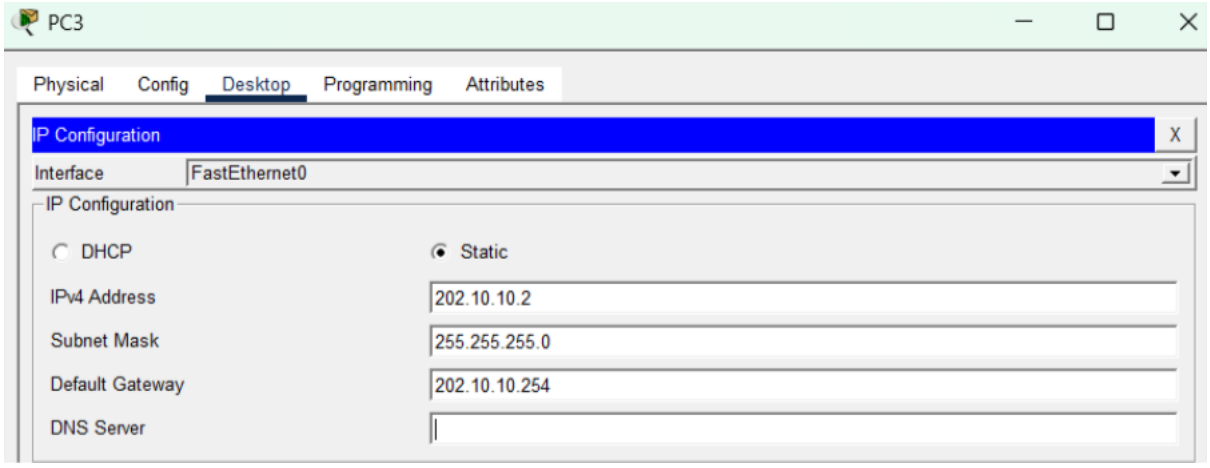
For PC 1



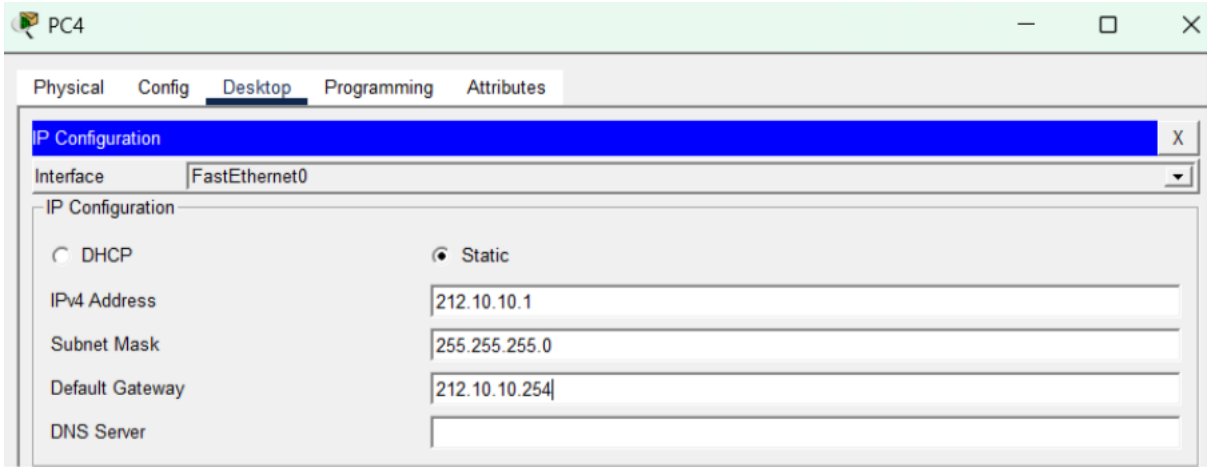
For PC 2



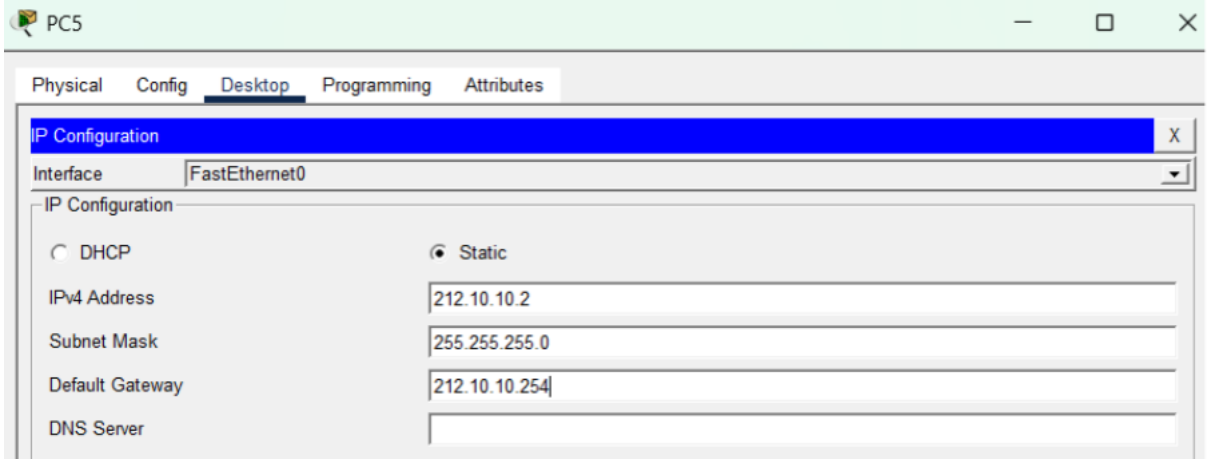
For PC 3



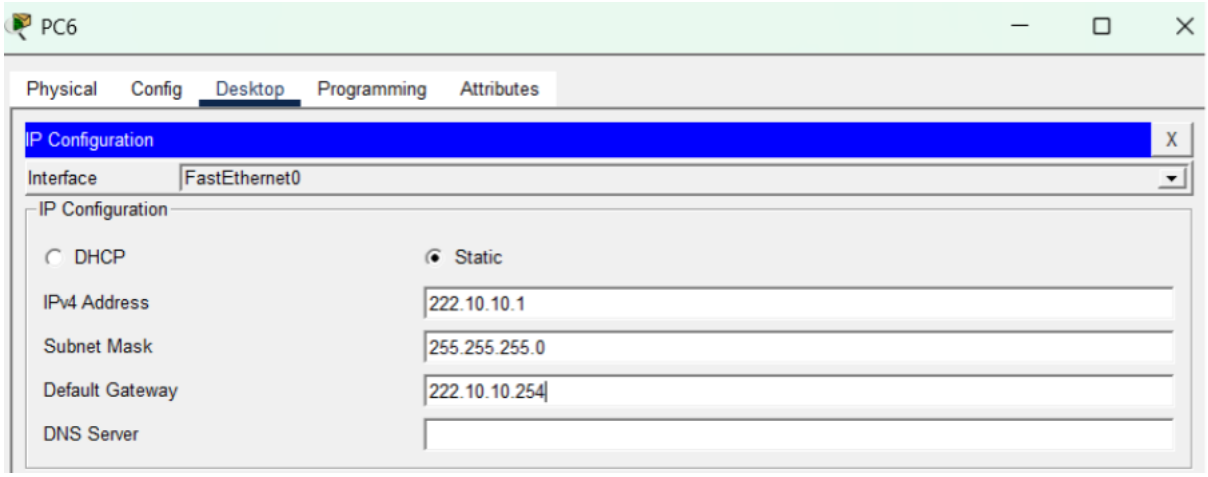
For PC4



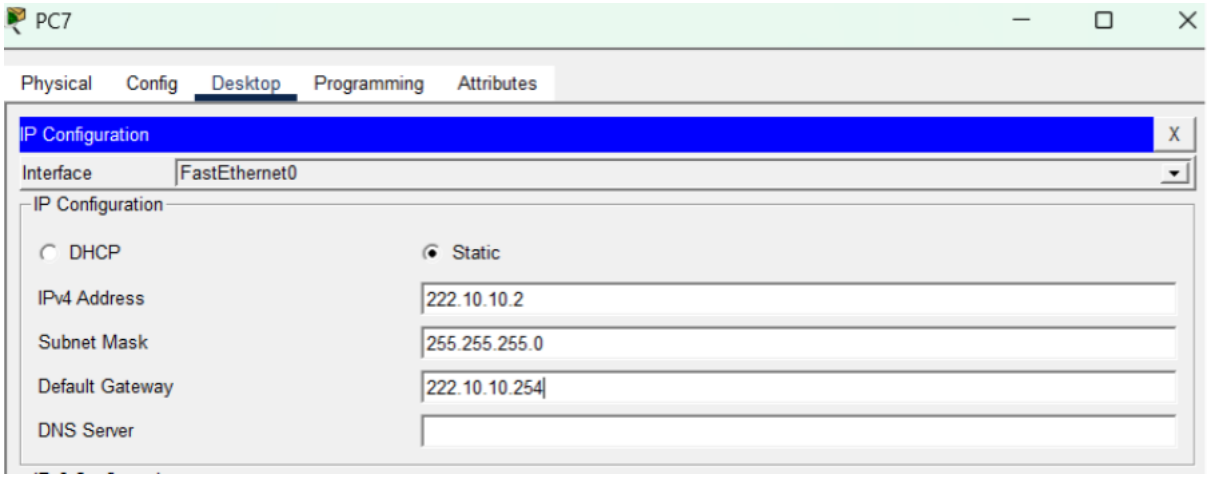
For PC5



For PC6


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
222.10.10.1
Subnet Mask
255.255.255.0
Default Gateway
222.10.10.254
DNS Server

For PC7


PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
222.10.10.2
Subnet Mask
255.255.255.0
Default Gateway
222.10.10.254
DNS Server